COM S 2300 HW5

1a) **Prove:** If f ○ g is injective, then g is one-to-one.

Counter example:

Let us assume that:

A= {3,4,5}

B={a,b}

C={0,1}

Now, we can define the function g: A → B as:

g(3) = a, g(4) = a and g(5) = b

Now lets define the function f: B → C

f(a) = 0 , f(b) = 1

Now, compute f o g:

(f og)(3)=f(g(3))=f(a)=0,
(f o g)(4)=f(g(4))=f(a)=0,
(f o g)(5)=f(g(5))=f(b)=1.

Since f is one-one, which implies g(a1) = g(a2) and a1 = a2 , so f o g is one-to-one, however, we cannot say anything about g so it is not one-to-one

b) **Counter - example**

Let us assume that:

A = {a,b}
B = {1,2,3}
C = {x,y}

So here we can say that:

g(a) = 1
g(b) = 2

f(g(a)) =f(1) = x
f(g(b)) = f(2) = x
f(3) = y

Therefore, we can say that f ○ g is onto since for all elements in C, there exists some elements in A, which makes f ○ g is onto C,however, there are some elements in B(e.g: 3) for which there does not exist any element in A, which means that g is not onto B.

Hence, g is not onto, if f ○ g is onto.

2 a) **Reflexive:** A relation is reflexive if (x,x) ∈ R1.

The relation is not reflexive because if we take x = 1, then we have 1*1=1 that is less than 3.

Anti-reflexive: A relation is said to be anti-reflexive if (x,x) does not belong to R1 and x*x >=3.

The relation is not anti-reflexive because, for instance, if we take x = √3, then √3*√3 = 3 which is equal to and 3 and satisfies the equation x*x>=3, which means √3 is a rational number, however it is not a rational number.

Hence, the relation is not anti-reflexive.

Symmetric: A relation is said to be symmetric if {x,y} ∈ R1 and {y,x} ∈ R2, that is, both xy and yx are greater than or equal to 3.

If we take x=4 and y=2 then we get xy=8 and yx = 8 as well. Since,the outcomes of both of these products is greater than 3 and both 2 and 4 are real numbers, we can say that this relation is symmetric.

Anti- Symmetric: A relation is said to be symmetric if (x,y) ∈ R1 and (x,y) ∈ R2, implies, x = y

Let us assume that x =1 and y = 4, since 1*4 = 4 that is greater than that means 1 and 4 are real numbers, however, 1 is not equal to 4, so that means the relation is not anti-symmetric.

Hence proved, the relation is anti - symmetric.

Transitive: A relation can be transitive if (x,y) ∈ R1, (y,z) ∈ R1, implies(x,z) ∈ R1.

Let us assume that x = 1, y = 4 and z = 1, so xy = 4, yz = 4 and xz = 1. Since the respective of products of xy and yz are greater than 3, so we can say that (x,y) and (y,z) belong to R1, however, since xz=1 that is less than 3, we can that (x,z) does not belong to R1,so the relation is not transitive.

Hence proved.

Therefore, we can say that R1 is symmetric.

b) **Reflexive:** The relation is not reflexive because it's not true when x=1,2,3,4…., it only holds true when x = 0, since at that time 0=4(0) that is 0.

Anti-reflexive: Since this equation holds true for when x = 0, we can say that there exists a reflexive pair for which this condition holds true, so it's not anti-reflexive.

Symmetric: If this relation is symmetric then it must follow $(x,y) \in R2 \Rightarrow (y,x) \in R2$. So, if x=4y, then y must be equal to 4x as well.

$$x = 4y \ (1)$$
$$y = 4x \ (2)$$
$$y = 4(4y) \text{ (substituting x from equation 1)}$$
$$y = 16y$$

Hence, we found out that after plugging in the value of x in y = 4x, we can get that y = 16y, which is not true for every rational number.

Thus, we can say that the relation is not symmetric.

Anti-Symmetric: To prove that this relation is anti-symmetric, we must prove that if $(x,y) \in R2 \Rightarrow (y,x) \in R2$, then x = y. So, over here we 2 equations:

$$x = 4y \ (1)$$
$$y = 4x \ (2)$$

Substituting x in equation 2:

$$y = 16y$$
$$0 = 16y - y$$
$$0 = 15y$$

Since we got 15y = 0, we can say that y = 0 and x = 0 from equations (1) and (2). So, as we can see that there exists one case where x and y are equal and belong to R2, we can say that the relation is not anti-symmetric.

Transitive: In order to prove that the relation is transitive, we need to show that if $(x,y) \in R2$, $(y,z) \in R2$, then $(x,z) \in R2$.So, over here we 2 equations:

$$x = 4y \ (1)$$

$$y = 4z \ (2)$$

Substituting y in equation 1:

$$x = 4(4z)$$
$$x = 16z$$

Since x = 16z and not 4z, we can say that the relation is not transitive.

Thus, over here the relation is anti-symmetric.

3) **Reflexive:** A relation is reflexive if $(x,x) \in R3$.

The relation is reflexive because when we divide x by x we get the quotient(answer) as 1, which is an integer. Thus, we can say that this relation is reflexive, for all positive real numbers.

Anti-Symmetric: To prove that this relation is anti-symmetric, we must prove that if $(x,y) \in R3 \Rightarrow (y,x) \in R3$, then x = y. So, over here we 2 equations:

$$x/y = k, \text{ for some integer } k$$
$$y/x = m, \text{ for some integer } m.$$

So we have:

$$x=ky \ (1)$$
$$y = mx \ (2)$$

Substitute the value of x in y:

$$y = m(ky)$$
$$y = kmy$$
$$1 = km$$

So, over here we can have 2 cases, k = m = 1 and k = m = -1. Since x and y are positive real numbers the 2nd case is definitely not possible.

Let us take a look at the 1st case, since k = m = 1, so after plugging in the values of both the constants in equations (1) and (2), we get that x = y.

Hence, the relation is anti-symmetric.

Transitive: In order to prove that the relation is transitive, we need to show that if (x,y) ∈ R3, (y,z) ∈ R3, then (x,z) ∈ R3.So, over here we 2 equations:

$$x/y = k, \text{ for some integer } k \ (1)$$
$$y/z= m, \text{ for some integer } m \ (2)$$

So:

$$y = mz$$

Substitute the value of y in (1):

$$x/mz = k$$
$$x/z = mk$$

Since x/z = mk, where m and k, both are integers,we can say that (x,z) ∈ R3, so this relation is transitive. Hence, R3 is reflexive, anti-symmetric and transitive.

4) **Reflexive:** A relation is reflexive if (x,x) ∈ R4

Over here if assume that n = 0, then we can say that:

$$x = x + 0/2 = x$$

Hence, we can say that there exists a pair (x,x) ∈ R4 as it satisfies the equation above when x = 0.

Symmetric: A relation is symmetric if (x,y) ∈ R4, (y,x) ∈ R4 as well.

Over here we the equation:

$$x = y + (n/2) \ (1)$$

After rearranging it we get:

$$y = x - (n/2)$$

Which can be written as:

$$y = x + (-n/2) \ (2)$$

In (2) we can see that -n is also an integer, so (2) must satisfy (1) and thus we can say that (y,x) ∈ R4.

Hence, we can say that the relation is symmetric.

Transitive: A relation is transitive if $(x,y) \in R4$, $(y,z) \in R4$, then $(x,z) \in R4$ is proven to be true.

Since we assume that $(x,y) \in R4$ and $(y,z) \in R4$ as, then we can say that:

$$x= y + n/2 \ (1), \text{ for some integer } n$$
$$y = z + m/2 \ (2), \text{ for some integer } m$$

Substituting the value of y from (2) in (1):

$$x= z + (m+n)/2 \ (3), \text{ for some integers } m \text{ and } n$$

Since m and n are integers we know that the sum of the integers is also an integer, so we can say that (3) satisfies the given equation $x = y + n/2$ (where n is an integer), so $(x,z) \in R4$.

Hence the relation is transitive.

The equivalence class of 2 can be written as:

$$y = 2 + n/2, \text{ for some integer } n.$$
$$[2] = \{ y \in Z| \ y = 2 + n/2, \text{ for some integer } n\}$$

So its set builder notation can be written as:

$$\{...,-2.5, -2, 1.5,-1,-0.5,0,0.5,1,1.5….\}$$

Similarly equivalence class of y can be written as:

$$y= \pi + n/2$$
$$[\pi] = \{y \in Z| \ y = \pi + n/2, \text{ for some integer } n\}$$

So it’s set builder notation can be written as:

$$\{....,\pi-1,\pi-0.5,\pi+0.5,\pi+1,.....\}$$

5a) **Reflexive:** The relation will be reflexive if (a,b) if we prove that $((a,b),(a,b)) \in R5$, where $(a,b) \in Z^+ \times Z^+$.

Since $a*b = a*b$, we can say that the $((a,b),(a,b)) \in R5$ and the relation is reflexive.

Symmetric: The relation will be symmetric if $((a,b),(c,d)) \in R5$, then $((c,d),(a,b)) \in R5$.

Since $((a,b),(c,d)) \in R5$, so $ad = cb$, and so we can say that $cb = ad$, and hence we can say that $((c,d),(a,b)) \in R5$.

Therefore, this relation is symmetric.

Transitive: The relation will transitive if $((a,b),(c,d)) \in R5$, $((c,d), (x,y)) \in R5$, then $((a,b),(x,y)) \in R5$.

Since $((a,b),(c,d)) \in R5$ and $((c,d), (x,y)) \in R5$, then we can say that $ad = cb$ (1)and $cy = dx$ (2)

Now, multiply both the sides of (1) by y.

$$ady = cby$$

Now, multiply both the sides of (2) by b.

$$bcy = bdx$$

Now, since $bcy = ady$

$$ady = bdx$$
$$ay = bx$$

Therefore, we can say that $((a,b),(x,y)) \in R5$ and hence we can say that the relation is an equivalence relation.

b) If $((a,b),(c,d)) \in R5$ then $ad = bc$ and when that happens the 2 functions become equal, so the relation R5 groups the pairs together if their fractions are equal. So we can say that:

$$a/b = c/d$$

Thus f can only be:

$$f(a,b) = a/b$$

Here we can say that each function assigns its value to the simplest fraction, which ensures that equivalent pairs have the same function values.

c) The equivalence class consists of the all the pairs such that:

$$a/b = 1$$
$$a = b$$

Thus the equivalence class of (1,1) can be defined as:

$$\{(a,b) \mid a = b,\ a,b \in Z^+\}$$

d) Since 2 pairs (a,b) and (c,d) are in the same equivalence class, so we can say that:

$$a/b = c/d$$

As each equivalence class corresponds to a unique rational number, so we can represent the set of the equivalent class as:

$$\{(a,b) \mid a/b = r, \text{ where r is any positive rational number.}\}$$

Now, from this set we can represent the equivalence class as:

$$[(a,b)] = \{(c,d) \in Z^+ \times Z^+ \mid ad = bc\}$$

By looking at this class, we can say that each pair in this class can be represented in the form of a/b.

6a) **Reflexive:** A relation is reflexive if $(f,f) \in R6$, for all f.

Since, for any function f, $f(0) = f(0)$, we can say that the relation is reflexive.

Symmetric: A relation is said to be symmetric if $(f,g) \in R6$, then$(g,f) \in R6$ as well.

Since $(f,g) \in R6$,we can say that $f(0) = g(0)$. Now, since the values of these functions are equal at $x = 0$, then we can also say that $g(0) = f(0)$ and hence $(g,f) \in R6$ and this makes the relation symmetric.

Transitive: A relation is said to be transitive if $(f,g) \in R6$, $(g,h) \in R6$, then $(f,h) \in R6$.

Since $(f,g) \in R6$ and $(g,h) \in R6$, then we can say that $f(0) = g(0)$ and $g(0) = h(0)$, and because of that we can say $f(0) = g(0)$, since both of them are equal to $g(0)$ so they have to be equal. Hence, the relation is transitive.

Hence the relation is an equivalence relation.

Now, we know that in an equivalence class a function f will contain all the functions where f(0) and g(0) are going to be equal. So that means the function will contain:

$$f = \{g \mid g(0) = f(0)\}$$

Thus, its equivalence class can be written as:

$$[f] = \{g \in Z^Z \mid g(0) = f(0)\}$$

b) **Reflexive:** A relation is reflexive if (f,f) $\in$ R7, for all f.

If we assume C to be 0, then we know that f(x)-f(x) = 0 = C, so we can say that the relation is reflexive.

Symmetric: A relation is said to be symmetric if (f,g) $\in$ R7, then(g,f) $\in$ R7 as well.

Since (f,g) $\in$ R7,we can say that f(x) - g(x) = C . Now, we can say that, since f(x) - g(x) = C then g(x) - f(x) = -C, which is also an integer and satisfies the relation that says the difference between 2 functions must be equal and this makes the relation symmetric.

Transitive: A relation is said to be transitive if (f,g) $\in$ R7, (g,h) $\in$ R7, then (f,h) $\in$ R7.

Since (f,g) $\in$ R7 and (g,h) $\in$ R7, then we can say that:

$$f(x) - g(x) = C1, \text{ for some integer } C1 \ (1)$$
$$g(x) - h(x) = C2, \text{ for some integer } C2 \ (2)$$

Add (1) and (2)

$$f(x) - g(x) + g(x) - h(x) = C1 + C2$$
$$f(x) - h(x) = C1 + C2, \text{ for some integers } C1 \text{ and } C2$$

Since the sum of 2 integers is always an integer we can say that f(x) - h(x) = a, where a = C1+ C2. Thus, the relation is transitive and it shows an equivalence relation.

Here we know that an equivalence class of f will contain all the functions for whom, the difference between them is always constant. Thus an equivalence class of f can be written as:

$$[f] = \{g \in Z^\wedge Z \mid g(x) = f(x) + C, \text{ where } C \in Z\}$$

$$[f] = \{g \in Z^\wedge Z \mid g(0) = f(0)\}$$